NAME (PRINT): ____
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STUDENT #: ______________________ **SIGNATURE:** ____________________

UNIVERSITY OF TORONTO MISSISSAUGA
APRIL 2022 FINAL EXAMINATION
STA457H5S
Applied Time Series Analysis
Masoud Ataei
Duration - 3 hours
Aids: Non-Programmable Calculators; 1 page(s) of double-sided Letter (8-1/2 x 11) sheet

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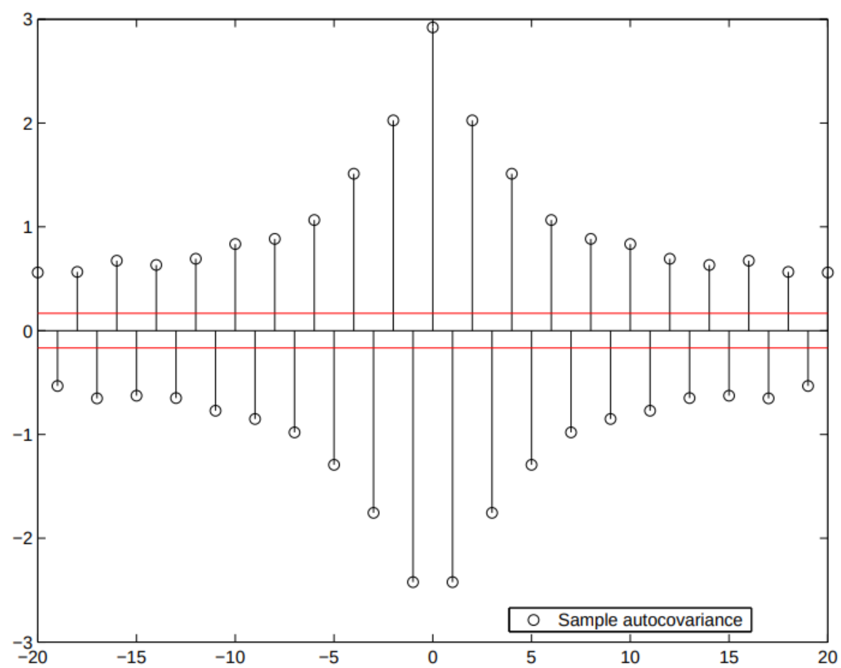
*Please note, once this exam has begun, you **CANNOT** re-write it.*

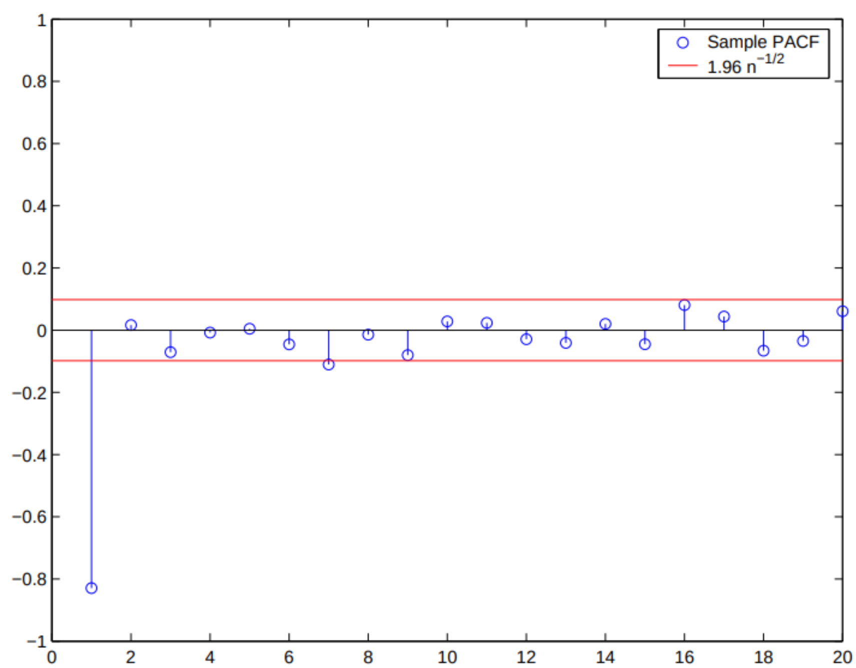
INSTRUCTIONS and POLICIES:

- Please DO NOT rip off any page from this booklet.
- For all questions, complete solutions are required. Show your work to earn full marks and then circle the final answer. Correct answers with no justifications will not receive any marks.
- Simplify final answers and round to 3 decimal places where appropriate.

Question	**1**	**2**	**3**	**4**	**5**	**6**	**TOTAL**
Value	10	15	15	20	20	20	**100**
Points							

Question 1 (10 points). The following plots depict the ***sample autocovariance*** and the ***sample partial autocorrelation*** functions based on a realization of size n from a ***stationary*** stochastic process X_t.





a) Should one employ an AR(p) or MA(q) to model the time series data realized from this process? What would the order of the model then be? **Please fully justify your answer.**

b) Use the given plots to find the $\mathrm{Var}(X_t)$?

Question 2 (15 points). Consider the following MA(2) process

$$X_t = W_t + \theta_1 W_{t-1} + \theta_2 W_{t-2},$$

where $W_t \overset{iid}{\sim} w_n(0, \sigma_w^2)$. Prove that

$$|\rho_2| \leq \frac{1}{2}.$$

Question 2 (cont'd).

Question 2 (cont'd).

Question 3 (15 points). Let $\{X_t\}$ be a stochastic process given by

$$X_t = A\sin(2\pi t + \theta), \qquad t = 0, \pm 1, \pm 2, \ldots,$$

where A is a random variable with zero mean and unit variance, while θ is a constant. Determine whether this process is (weakly) stationary or not.

Question 3 (cont'd).

Question 3 (cont'd).

Question 4 (20 points). Let $\{X_t\}$ be a stochastic process defined by

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + W_t,$$

where $\{W_t\} \sim wn(0, \sigma_w^2)$. Based on a realization of size $n = 100$ from this process we obtain the following ***sample autocorrelation coefficients***:

$$r_1 = 0.7\,, \qquad r_2 = 0.4\,, \qquad r_3 = 0.3\,.$$

Estimate ϕ_1 and ϕ_2 using the *Method of Moments*. You may assume that $\mathbb{E}[X_t] = 0$.

Question 4 (cont'd).

Question 4 (cont'd).

Question 5 (20 points). Let $\{Y_t\}$ be a stationary process with mean zero and let a and b be constants. Also, consider the following processes

$$X_t = a + bt + S_t + Y_t,$$

and

$$Z_t = \nabla \nabla_{12} X_t = (1 - B)(1 - B^{12}) X_t,$$

where S_t is a seasonal component with period 12. Derive the ***autocovariance function*** of $\{Z_t\}$ in terms of the ***autocovariance function*** of $\{Y_t\}$, and then show that $\{Z_t\}$ is a stationary process.

Question 5 (cont'd).

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Question 5 (cont'd).

Question 6 (20 points). Consider the following process

$$X_t = 0.2X_{t-1} - 0.2X_{t-2} + 0.6X_{t-3} + W_t,$$

where $\{W_t\} \sim N(0, \sigma_w^2)$. The following values have been observed for this time series:

$$x_1 = 3.1, \quad x_2 = 0.93, \quad x_3 = -2.3, \quad x_4 = 3.2, \quad x_5 = 1.3,$$

$$x_6 = -2.5, \quad x_7 = 1.6, \quad x_8 = 3, \quad x_9 = -3.5, \quad x_{10} = -0.74.$$

Use the best linear predictor to compute the one-step ahead forecast $\hat{x}_t(1)$ and give its 95% prediction interval.

Question 6 (cont'd).

Question 6 (cont'd).

Table entry for z is the area under the standard Normal curve to the left of z.

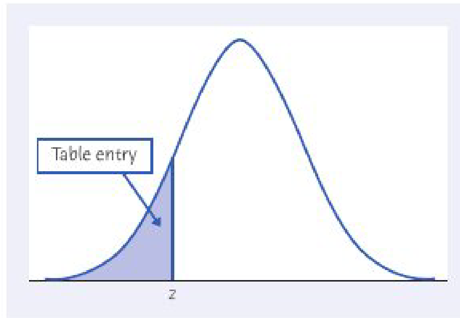

TABLE A Standard Normal cumulative proportions

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
−3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
−3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
−3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
−3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
−2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
−2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
−2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
−2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
−2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
−2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
−2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
−2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
−2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
−2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
−1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
−1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
−1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
−1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
−1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
−1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
−1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
−1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
−1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
−1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
−0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
−0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
−0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
−0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
−0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
−0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
−0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
−0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
−0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
−0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

Table entry for z is the area under the standard Normal curve to the left of z.

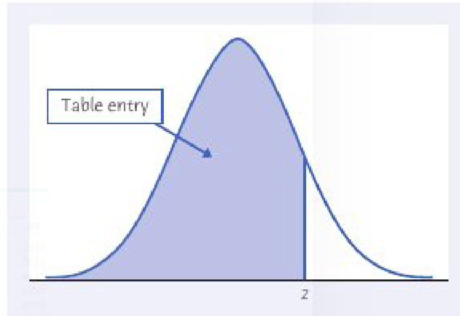

TABLE A Standard Normal cumulative proportions (*continued*)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

END OF EXAM*

* This page will not be graded unless you clearly indicate in the exam that something here is to be marked.